

PRAKTIKUM BASIS DATA

MODUL 9

Scripting Procedure dan Function



**Universitas
Telkom**

Oleh:

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103122430009

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

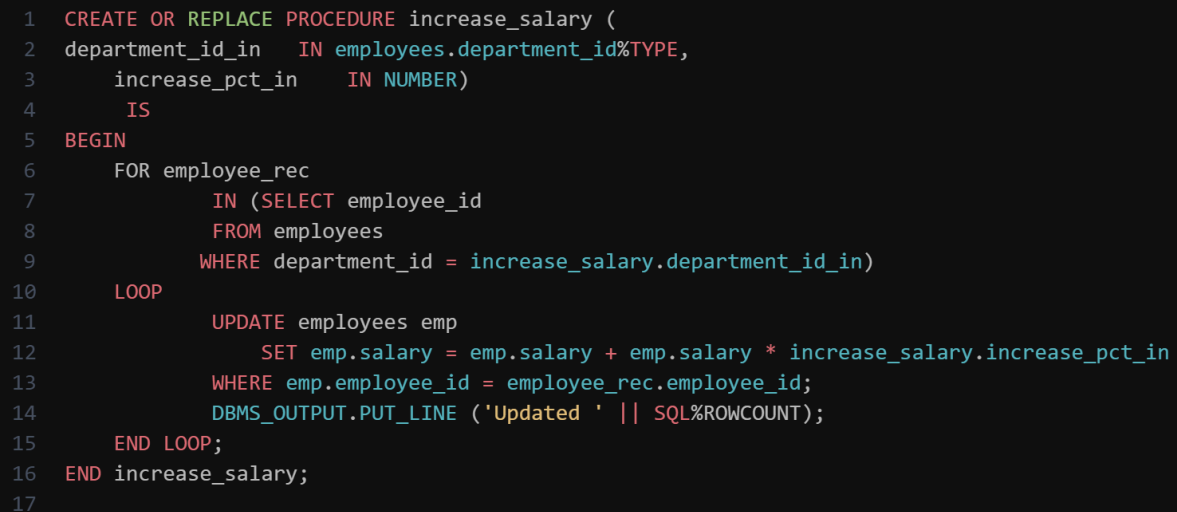
2026

MODUL 9

JURNAL

Running [code sql](#) berikut, screenshot hasilnya dan analisis hasil code tersebut

1. create table employees as select * from hremployees.
2. select * from employees.
- 3.



```
1 CREATE OR REPLACE PROCEDURE increase_salary (  
2 department_id_in IN employees.department_id%TYPE,  
3 increase_pct_in IN NUMBER)  
4 IS  
5 BEGIN  
6     FOR employee_rec  
7         IN (SELECT employee_id  
8             FROM employees  
9             WHERE department_id = increase_salary.department_id_in)  
10    LOOP  
11        UPDATE employees emp  
12            SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in  
13            WHERE emp.employee_id = employee_rec.employee_id;  
14        DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);  
15    END LOOP;  
16 END increase_salary;  
17
```

4. BEGIN
 increase_salary (90, .50);
 ROLLBACK;
END;
/
select * from employees where department_id = 90
5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.

Jawaban:

```
Oracle SQL Developer: JRNAL_Modu09_Basdat_03122430009
File Edit View Navigate Run Source Tools Window Help

Connections
Oracle Connections
+ DB_FLM
+ DB_FLM_Jurnal09_Basdat_03122430009
+ DB_FLM_Modu09
+ DB_FLM_Modu09_Basdat_03122430009
+ JRNAL_Modu02_Basdat_03122430009
+ JRNAL_Modu09_Basdat_03122430009
+ Modu03_Basdat_03122430009
+ Modu03_Basdat_Semester 4 SE08
+ Modu09_Basdat_03122430009
+ TP_Modu03_Basdat_03122430009
+ TP_Modu04_Basdat_03122430009
+ TP_Modu05_Basdat_03122430009
+ TP_Modu06_Basdat_03122430009
+ TP_Modu07_Basdat_03122430009
Database Schema Service Connectors

Worksheet: Query Builder
CREATE TABLE hrsemployees (
  employee_id NUMBER(6) NOT NULL,
  first_name VARCHAR2(20),
  last_name VARCHAR2(25) NOT NULL,
  email VARCHAR2(25) NOT NULL,
  phone_number VARCHAR2(20),
  hire_date DATE NOT NULL,
  job_id VARCHAR2(10) NOT NULL,
  salary NUMBER(5,2),
  commission_pct NUMBER(2,2),
  manager_id NUMBER(6),
  department_id NUMBER(4),
  CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);

INSERT INTO hrsemployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-2003', 'DD-MON-YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO hrsemployees VALUES (101, 'Neena', 'Kochhar', 'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005', 'DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO hrsemployees VALUES (102, 'Lex', 'De Haan', 'LDEHAAN', '515.123.4569', TO_DATE('13-JAN-2001', 'DD-MON-YYYY'), 'AD_VP', 17000, NULL, 100, 90);
INSERT INTO hrsemployees VALUES (103, 'Alexander', 'Russell', 'ARUSSEL', '590.420.4567', TO_DATE('03-JAN-2006', 'DD-MON-YYYY'), 'IT_PROG', 9000, NULL, 102, 60);
INSERT INTO hrsemployees VALUES (104, 'Bruce', 'Ernst', 'BERNST', '590.420.4568', TO_DATE('21-MAY-2007', 'DD-MON-YYYY'), 'IT_PROG', 6000, NULL, 103, 60);
INSERT INTO hrsemployees VALUES (107, 'Diana', 'Lorentz', 'DLORENTZ', '590.420.5567', TO_DATE('07-FEB-2007', 'DD-MON-YYYY'), 'IT_PROG', 4200, NULL, 103, 60);
INSERT INTO hrsemployees VALUES (124, 'Kevin', 'Mourgos', 'KMORGOS', '650.123.5234', TO_DATE('16-MAY-2007', 'DD-MON-YYYY'), 'ST_MAN', 5800, NULL, 100, 90);
INSERT INTO hrsemployees VALUES (141, 'Trenna', 'Raja', 'TRAJA', '650.121.8009', TO_DATE('17-OCT-2003', 'DD-MON-YYYY'), 'ST_CLERK', 3500, NULL, 124, 50);

PL/SQL procedure successfully completed.

SQL>
SQL> CREATE TABLE hrsemployees (
  2 employee_id NUMBER(6) NOT NULL,
  3 first_name VARCHAR2(20),
  4 last_name VARCHAR2(25) NOT NULL,
  5 email VARCHAR2(25) NOT NULL,
  6 phone_number VARCHAR2(20),
  7 hire_date DATE NOT NULL,
  8 job_id VARCHAR2(10) NOT NULL,
  9 salary NUMBER(5,2),
  10 commission_pct NUMBER(2,2),
  11 manager_id NUMBER(6),
  12 department_id NUMBER(4),
  13 CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
  14 );

Table HRSEMPLOYEES created.
```

```
Oracle SQL Developer: JRNAL_Modu09_Basdat_03122430009
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Connections
Oracle Connections
+ DB_FLM
+ DB_FLM_Jurnal09_Basdat_03122430009
+ DB_FLM_Modu09
+ DB_FLM_Modu09_Basdat_03122430009
+ JRNAL_Modu02_Basdat_03122430009
+ JRNAL_Modu09_Basdat_03122430009
+ Modu03_Basdat_03122430009
+ Modu03_Basdat_Semester 4 SE08
+ Modu09_Basdat_03122430009
+ TP_Modu03_Basdat_03122430009
+ TP_Modu04_Basdat_03122430009
+ TP_Modu05_Basdat_03122430009
+ TP_Modu06_Basdat_03122430009
+ TP_Modu07_Basdat_03122430009
Database Schema Service Connectors

Worksheet: Query Builder
manager_id NUMBER(6),
department_id NUMBER(4),
CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);

INSERT INTO hrsemployees VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', TO_DATE('17-JUN-2003', 'DD-MON-YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);
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INSERT INTO hrsemployees VALUES (124, 'Kevin', 'Mourgos', 'KMORGOS', '650.123.5234', TO_DATE('16-MAY-2007', 'DD-MON-YYYY'), 'ST_MAN', 5800, NULL, 100, 90);
INSERT INTO hrsemployees VALUES (141, 'Trenna', 'Raja', 'TRAJA', '650.121.8009', TO_DATE('17-OCT-2003', 'DD-MON-YYYY'), 'ST_CLERK', 3500, NULL, 124, 50);
INSERT INTO hrsemployees VALUES (149, 'Elion', 'Elael', 'EELAEEL', '011.44.1344.429010', TO_DATE('29-JAN-2006', 'DD-MON-YYYY'), 'SA_MAN', 10500, .2, 100, 80);
INSERT INTO hrsemployees VALUES (174, 'Ellen', 'Elael', 'EELAEEL', '011.44.1644.429267', TO_DATE('11-MAY-2004', 'DD-MON-YYYY'), 'SA_REP', 11000, .3, 149, 80);

COMMIT;

CREATE TABLE employees AS SELECT * FROM hrsemployees;

SELECT * FROM employees;

CREATE OR REPLACE PROCEDURE increase_salary (
  1 row inserted.

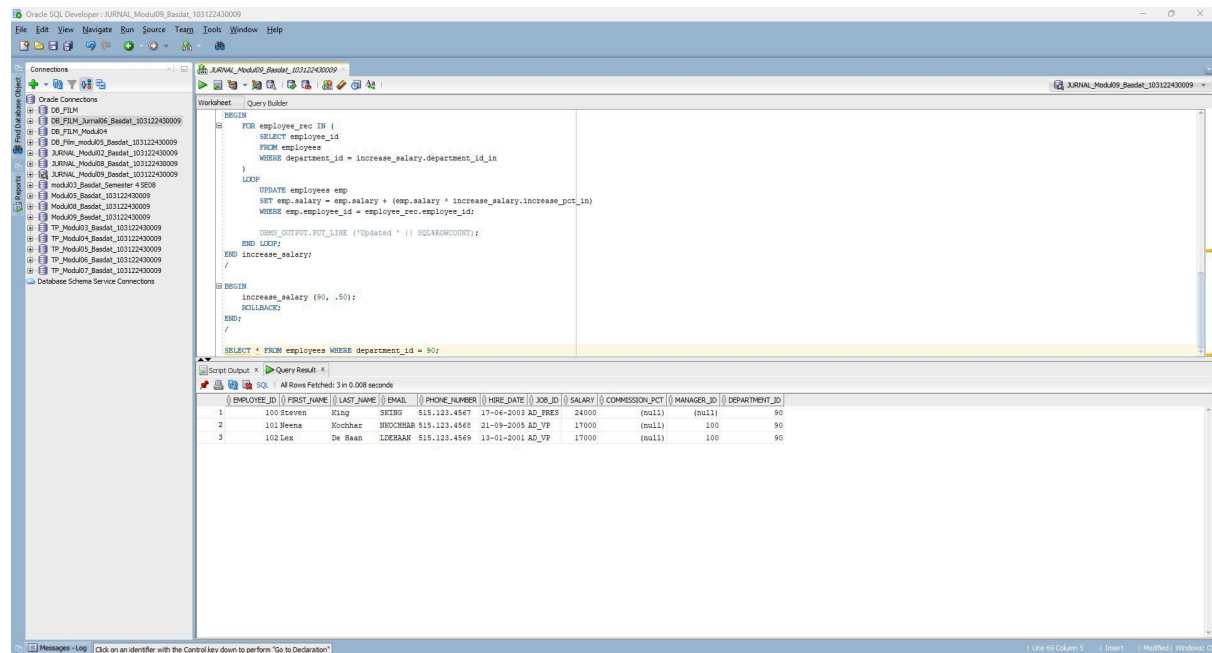
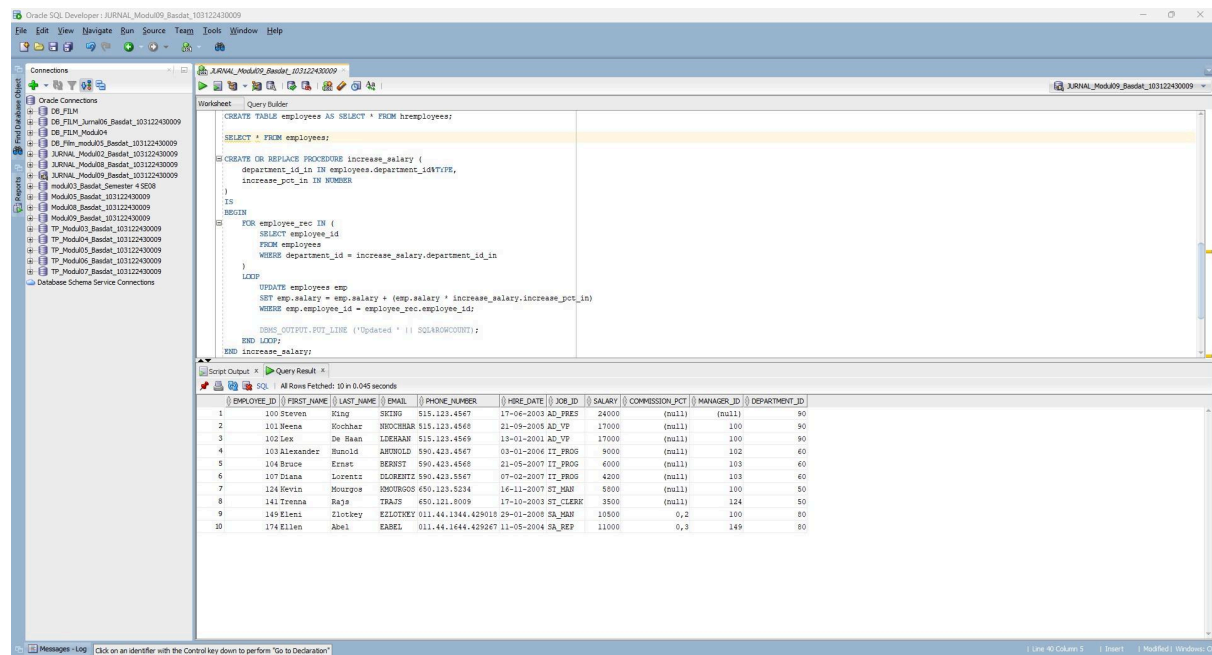
SQL> INSERT INTO hrsemployees VALUES (124, 'Kevin', 'Mourgos', 'KMORGOS', '650.123.5234', TO_DATE('16-MAY-2007', 'DD-MON-YYYY'), 'ST_MAN', 5800, NULL, 100, 90);
1 row inserted.

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1 row inserted.

SQL> COMMIT;
Commit complete.
```



5. Jika statement `ROLLBACK`; pada blok kode nomor 4 dihapus, maka perubahan kenaikan gaji sebesar 50% akan disimpan secara permanen di dalam database. karena terdapat perintah `ROLLBACK`; segala perubahan gaji yang dilakukan oleh procedure akan dibatalkan/dikembalikan ke kondisi semula. Hal ini dibuktikan pada hasil `SELECT * FROM employees WHERE department_id = 90`; di akhir kode, di mana gaji Steven King tetap 24000 dan Neena Kochhar tetap 17000. Jika `ROLLBACK`; dihapus , maka saat tabel di-`SELECT` lagi, gaji Steven King akan berubah permanen menjadi 36000 dan Neena Kochhar menjadi 25500.